Algebraic Coding Theory

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Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill book [1] and Lint book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to $LaTeX$.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

Contents

1 Vector spaces over finite fields, quick recap

Let q be a prime power.

$V(n,q)$: set $GF(q)^n$ of all ordered n-tuples over $GF(q)$. $V(n,q)$ is a vector space.

A set of vectors $\{v_1, v_2, \ldots, v_r\}$ is said to be *linear dependent* if $\exists$ scalars $a_1, \ldots, a_r$, not all zero, such that

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0$$

They are *linear independent* if

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0 \overset{implies}{\Longrightarrow} a_1 = a_2 = \ldots = a_r = 0$$

Let C a subspace of $V(n,q)$. Then a subset $\{v_1, v_2, \ldots, v_r\} \subseteq C$ is called a *generating set* (or *spanning set*) of C if every vector in C can be expressed as a linear combination of $v_1, \ldots, v_r$.

A generating set of C which is also linearly independent is called a *basis* of C.

Theorem 4.3. Suppose $\{v_1, v_2, \ldots, v_k\}$ is a basis of a subspace C of $V(n,q)$. Then

i. every vector of C can be expressed uniquely as a linear combination of the basis vectors.

ii. C contains exactly q^k vectors.

Proof. i. suppose vector $x \in C$ such that

$$x = a_1v_1 + a_2v_2 + \cdots + a_kv_k$$

$$x = b_1v_1 + b_2v_2 + \cdots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \cdots + (a_k - b_k)v_k = 0.$$

But the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \ldots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \ldots, k.$$

ii. By (i), the q^k vectors

$$\sum_{i=1}^{k} a_i v_i \qquad (a_i \in \mathrm{GF}(q))$$

are precisely the distinct vectors of C.

□

2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:
A q-ary linear code $[n, k, d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

2.1 Some definitions

Let q be a prime power. View $(F_q)^n$ as the vector space $V(n, q)$.

Definition (Linear code over GF(q))**.** A *linear code over* $GF(q)$ is a subspace of $V(n, q)$, with $n \geq 0$.

Thus $C \subseteq V(n, q)$ is a linear code iff

i. $u + v \in C \;\; \forall u, v \in C$

ii. $au \in C \;\; \forall u \in C, \; a \in GF(q)$

If C is a k-dimensional subspace of $V(n, q)$, then the linear code C is called an $[n, k]$-code. (sometimes with minimum distance d: $[n, k, d]$-code).

Definition (Weight)**.** Weight, $w(x)$, of a vector $x \in V(n, q)$ is the number of non-zero entries of x.

Lemma 5.1. if $x, y \in V(n, q)$, then $d(x, y) = w(x - y)$.

Proof. the vector $x - y$ has non-zero entries in preceisely those places where x and y differ. □

Theorem 5.2. let C a linear code, let $w(C)$ be the smallest of weights of the non-zero codewords of C. Then $d(C) = w(C)$.

Proof. $\exists\ x, y \in C$ s.th. $d(C) = d(x, y)$.

By Lemma 5.1, $d(C) = w(x - y) \geq w(C)$, since $x - y$ is a codeword of the linear code C (ie. $x - y \in C$).

On the other hand, for $x \in C$,
$w(C) = w(x) = d(x, 0) \geq d(C)$, since $0 \in C$.

Hence, $d(C) \geq w(C)$ and $w(C) \geq d(C) \implies d(C) = w(C)$. □

Definition (Generator matrix (GM))**.** a $k \times n$ matrix whose rows form a basis of a linear $[n, k]$-code is called a *generator matrix* (GM) of the code.

2.2 Equivalence

Theorem 5.4. two known $k \times n$ matrices generate equivalent linear $[n, k]$-codes over $GF(q)$ if one matrix can be obtained from the other by a sequence of operations of the following types:

R1. permutation of the rows

R2. multiplication of a row by a non-zero scalar

R3. addition of a scalar multiple of one row to another

C1. permutation of the columns

C2. multiplication of any column by a non-zero scalar

Theorem 5.5. let G a GM of an $[n, k]$-code. Then by performing the previous operations, G can be transformed to the standard form $[I_k|A]$ where I_k is the $k \times k$ identity matrix, and A is a $k \times (n - k)$ matrix.

2.3 Encoding with a linear code

Let C a $[n, k]$-code over $GF(q)$ with GM G. C contains q^k codewords $\Longrightarrow$ can be used to communicate any one of q^k distinct messages.

Iddentify these messages with the q^k tuples of $V(k, q)$.

Encoding:
msg vector: $u = u_1 u_2 \dots u_k$; rows of G: $r_1 r_2 \dots r_k$. Encode

$$uG = \sum_{i=1}^{k} u_i r_i \in C$$

ie. uG is a codeword of C; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k, q) &\longrightarrow C \subseteq V(n, q) \\ u &\longmapsto uG \end{aligned}$$

If G is in standard form (5.5), $G = [I_k|A]$ with $A = [a_{ij}]$, then encoding u is:

$$x = uG = x_1x_2 \dots x_kx_{k+1} \dots x_n$$

where

$$x_i = u_i \text{ for } 1 \leq i \leq k, \text{ message digits}$$
$$x_{k+i} = \sum_{j=1}^{k} a_jiu_j \text{ for } 1 \leq i \leq n-k, \text{ check digits}$$

the check digits add redundancy to gie protection against noise.

2.4 Decoding with a linear code

Codeword $x = x_1x_2 \dots x_n$, received vector $y = y_1y_2 \dots y_n$, error vector $e = y - x = e_1e_1 \dots e_n$.

2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

Definition (Coset)**.** Let C a $[n,k]$-code over $GF(q)$ and $a \in V(n,q)$. Then the set $a + C = \{a + x | x \in C\}$ is called a *coset* of C.

Definition (Coset leader)**.** the vector having minimum weight in a coset is called the *coset leader*.

Lemma 6.3. Let $a + C$ a coset of C and $b \in a + C$. Then $b + C = a + C$.

Proof. since $b \in a + C \implies b = a + x$ for some $x \in C$. If $b + y \in b + C$ then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if $a + x \in a + C$ then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence $a + C \subseteq b + C$.

Therefore $b + C = a + C$. □

Theorem 6.4 (Lagrange)**.** suppose $C \subseteq V(n,q)$ is a $[n,k]$-code over $GF(q)$. Then

i. evvery vector of $V(n,q)$ is in some coset of C

ii. every coset contains exactly q^k vectors

iii. two cosets either are disjoint or coincide (partial overlap is impossible)

Proof. i. if $a \in V(n, q)$ then $a = a + 0 \in a + C$

ii. the map

$$\begin{aligned} C &\longrightarrow a + C \ \forall\ x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence $|a + C| = |C| = q^k$.

iii. by contradiction, suppose cosets $a+C,\ b+C$ overlap. Then for some vector v,

$$v \in (a + C) \cap (b + C)$$

thus for some $x, y \in C$, $v = a + x = b + y$.

Hence $b = a + (x - y) \in a + C$, thus by Lemma 6.3 we have $b + C = a + C$.

□

Corollary . 6.4 shows that $V(n, q)$ is partitioned into disjoint cosets of C:

$$V(n, q) = (0 + C) \cup (a_1 + C) \cup \ldots \cup (a_s + C)$$

where $s = q^{n-k} - 1$, and by Lemma 6.3 we may take $0, a_1, \ldots, a_s$ to be coset leaders.

Example 6.5. Let C be the binary $[4, 2]$-code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie. $C = \{0000,\ 1011,\ 0101,\ 1110\}$.
Then the cosets of C are

$$\begin{aligned} &0000 + C = C \text{ itself} \\ &1000 + C = \{1000,\ 0011,\ 1101,\ 0110\} \\ &0100 + C = \{0100,\ 1111,\ 0001,\ 1010\} = 0001 + C \\ &0010 + C = \{0010,\ 1001,\ 0111,\ 1100\} \end{aligned}$$

by Lemma 6.3: $0001 \in 0100 + C$, thus $0001 + C = 0100 + C$.

2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let p be the symbol error probability.

Probability that the error vector is agiven vector of weight i is $p^i(1-p)n - i$.

Theorem 6.7. Let C a binary $[n,k]$-code, and for $i = 0, 1, \ldots, n$ let α_i denote the number of coset leaders of weight i.

Then the probability $P_{corr}(C)$ that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Example 6.8. $[n,k]$-code from 6.5, coset leaders are $0000, 1000, 0100, 0010$. Hence $\alpha_0 = 1,\ \alpha_1 = 3,\ \alpha_2 = \alpha_3 = \alpha_4 = 0$, so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

Remark 6.9. if $d(C) = 2t+1$ or $d(C) = 2t+2$, then C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

Definition (Rate)**.** $[n,k]$-code C uses n symbols to send k message symbols; it has *rate* $R(C) = k/n$.

A good code will have a high rate.

Definition (Capacity)**.** the *capacity* $\mathscr{C}(p)$ of a binary symmetric channel with symbol error probability p is

$$\mathscr{C}(p) = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p)$$

Theorem 6.12 (Shanon's theorem)**.** channel binary symmetric with symbol error probability p.

Let $R < \mathscr{C}(p)$.

Then for any $\epsilon > 0$, $\exists$ for sufficient large n, an $[n,k]$-code C of rate $k/n \geq R$ such that $P_{err}(C) < \epsilon$.

Definition (P_{symb})**.** P_{symb} denotes the average probability that a message symbol is in error after decoding.

Theorem 6.14. let C a binary [n,k]-code and let A_i denote the number of codewords of C of weight i.

Then, if C is used for error detection, the probability of an incorrect message being undetected is

$$P_{undetec}(C) = \sum_{i=1}^{n} A_i \cdot p^i (1-p)^{n-i}$$

Example 6.15. for the code of 6.5,

$$P_{undetec}(C) = \underbrace{1 \cdot p^2(1-p)^2}_{0101} + \underbrace{2 \cdot p^3(1-p)^{4-3=1}}_{1011,\ 1110} = p^2 - p^4$$

2.6 Exercises

Exercise 5.1. is the binary (11, 24, 5)-code linear?

Proof. recall: a q-ary linear $[n, k, d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

$\implies$ $(11, 24, 5)$-code to be linear it would mean that $24 = q^k$.

Since it is binary, $q = 2$. But $24 \neq 2^k$, so it can not be linear. □

Exercise 5.2. E_n is the code of all even-weight vectors of $V(n, 2)$, which is linear. What are the parameters $[n, k, d]$ of E_n? Write a GM for E_n in standard form.

Proof.

$$E_n = \{x \in V(n, 2) | w(x) \text{ is even}\}$$

$\longrightarrow$ is the binary even-weight ode (the single parity-check code);
it's linear since

- the zero vector has even weight
- over $GF(2)$, the sum of two even-weight vectors has even weight

Recall: $F_q^n \iff V(n, q)$, so $V(n, 2) \cong F_2^n$

$$\implies E_n = \{(x_1, \ldots, x_n) \in F_2^n | x_1 + \ldots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in E_n
- words of weight 2 are in E_n

Therefore, parameters are $[n, n-1, 2]$
GM:

$$G = [I_{n-1}|1]$$

which generates vectors of the form

$$(a_1, \ldots, a_{n-1}, a_1 + \ldots + a_{n-1})$$

□

Exercise 5.5. Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

Proof. let C_e, C_o be subsets of C with even and odd weights respectively.

So that $C = C_e \cup C_o$ (disjoint union).

For binary vectors x, y:

$$w(x + y) = w(x) + w(y) \pmod 2$$

ie. over F_2, the parity of the weight is additive:

$$\begin{aligned} even + even &= even \\ even + odd &= odd \\ odd + odd &= even \end{aligned}$$

Now we have two possible cases:

- C has no odd weighted codewords:
  $C_0 = \emptyset,\ \forall x \in C\ x$ is even weighted

- C has at least one odd-weighted codeword:
  let $u \in C_o$, consider the map

  $$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

  with $c \in C_e,\ u \in C_o,\ \phi(c) \in C_o$.

  ϕ is bijective, since $\phi(\phi(c)) = (c + u) + u = c$,

  so ϕ is it's own inverse; hence $|C_e| = |C_o|$.

  Since $C = C_e \cup C_e$, half codewords are even weighted and half odd.

□

Exercise 6.1. construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Proof. recall: for a binary code with GM G, the code C consists of all binary linear combinations of the rows of G.

For G_1:
rows: $r_1 = (1, 0),\ r_2 = (0, 1))$. A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \quad \text{with } a, b \in \{0, 1\}$$

4 combinations:

$$\begin{aligned} 0 \cdot r_1 + 0 \cdot r_2 &= (0, 0) \\ 1 \cdot r_1 + 0 \cdot r_2 &= (1, 0) \\ 0 \cdot r_1 + 1 \cdot r_2 &= (0, 1) \\ 1 \cdot r_1 + 1 \cdot r_2 &= (1, 0) + (0, 1) = (1, 1) \end{aligned}$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

standard array: 00 01 10 11

For G_2:
rows: $r_1 = (1,0,1),\ r_2 = (0,1,1)$. Codeword: $a \cdot r_1 + b \cdot r_2$ with $a, b \in \{0,1\}$.
4 combinations:

$$\begin{aligned}
0 \cdot r_1 + 0 \cdot r_2 &= (0,0,0) \\
1 \cdot r_1 + 0 \cdot r_2 &= (1,0,1) \\
0 \cdot r_1 + 1 \cdot r_2 &= (0,1,1) \\
1 \cdot r_1 + 1 \cdot r_2 &= (1,0,1) + (0,1,1) = (1,1,0)
\end{aligned}$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$ so the 1st row of the standard array is: 000 101 011 110.
Now, pick a non used vector: 001, add it to the code C_2:

$$\begin{aligned}
&001 + C_{2_i} \\
&001 + 000 = 001 \\
&001 + 101 = 100 \\
&001 + 011 = 010 \\
&001 + 110 = 111
\end{aligned}$$

$\implies$ 2nd row of the standard array: 001 100 010 111
Thus the standard array is

$$\begin{aligned}
&000\ 101\ 011\ 110 \\
&001\ 100\ 010\ 111
\end{aligned}$$

ie. $2^3 = 8$ vectors, a $V(3,2)$ vector space. □

Exercise 6.4. C a binary $[n,k]$-code with minimum distance $2t+1$ (or $2t+2$). Given that p is very small, show that an approximate value of $P_{err}(C)$ is $\left(\binom{n}{t+1} - \alpha_{t+1}\right) p^{t+1}$, where α_{t+1} is the number of coset leaders of C of weight $t+1$.

Proof. From 6.9, if $d(C) = 2t+1$ (or $2t+2$) $Longrightarrow\ C$ can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \tag{1}$$

where α_i is the number of coset leaders of weight i

From the $\binom{n}{i}$ vectors of weight i, exactly α_i are coset leaders, so the non coset leaders are $\binom{n}{i} - \alpha_i$. This is when a decoding error happens.

$$\implies P_{err}(C) = \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9, $\forall i \leq t,\ \alpha_i = \binom{n}{i}$, thus $\binom{n}{i} - \alpha_i = 0$ for the first t terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left(\binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t+1} + \dots \end{aligned}$$

where the ... represent the terms with p^{t+2} and higher.

Then, since p is very small (by the exercise definition), $(1-p)^{n-t+1} \equiv 1$, and the other higher powers of p are negligible.

Hence,

$$P_{err}(C) = \left(\binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

3 Dual code, PCM and Syndrome

3.1 Dual code

Definition (Orthogonal)**.** Inner product:

$$u \cdot v = u_1 v_1 + \ldots + u_n \cdot v_n = \sum_{i=1}^{n} u_i v_i$$

If $u \cdot v = 0$, then u and v are called *orthogonal*.

Lemma 7.1. for any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

i. $u \cdot v = v \cdot u$

ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

Proof. (at exercise 7.1). □

Definition (Dual code)**.** given a linear $[n, k]$-code C, the *dual code* of C, denoted $C^\perp$, is defined to be the set of those vectors of $V(n, q)$ which are orthogonal to every codeword of C. ie.

$$C^\perp = \{v \in V(n, q) \mid v \cdot u = 0 \ \forall \ u \in C\}$$

Lemma 7.2. C a $[n, k]$-code with GM G. Then $v \in V(n, q)$ belongs to $C^\perp$ iff v is orthogonal to every row of G, ie.

$$v \in C^\perp \iff v \cdot G^T = 0$$

Proof. let rows of G be $r_1, \ldots, r_k$, suppose $v \cdot r_i = 0 \ \forall i$ (ie. v orthogonal to every row of G).

If $u \in C$ (ie. a codeword of C), then $u = \sum_{i=1}^{k} \lambda_i r_i$ for some scalars λ_i, so that

$$v \cdot u = v \cdot \sum_{i=1}^{k} \lambda_i r_i = \sum_{i=1}^{k} \lambda_i \cdot \underbrace{(v \cdot r_i)}_{0} = \sum \lambda_i \cdot 0 = 0$$

$\implies$ hence v is orthogonal to every codeword of C, and so is in $C^\perp$. □

Theorem 7.3. C a $[n, k]$-code over $GF(q)$. Then the dual code $C^\perp$ of C is a linear $[n, n-k]$-code.

Proof. i. first show that $C^\perp$ is a linear code:

let $v_1, v_2 \in C^\perp$ and $\lambda, \mu \in GF(q)$. Then $\forall u \in C$,

$$\begin{aligned}(\lambda v_1 + \mu v_2) \cdot u &= \lambda(v_1 \cdot u) + \mu(v_2 \cdot u) \\ &\text{notice that } v_1, v_2 \in C^\perp\text{, so} \\ &\text{that } v_1, v_2 \text{ are orthogonal with } u \\ &\text{hence } (v_1 \cdot u) \text{ and } (v_2 \cdot u) \text{ are } 0 \\ &= \lambda \cdot 0 + \mu \cdot 0 = 0\end{aligned}$$

Hence $\lambda \cdot v_1 + \mu \cdot v_2 \in C^\perp$, and so $C^\perp$ is linear, since it is closed under linear combinations.

ii. Next, show that $C^\perp$ has dimension $n-k$:

Let $G = [g_{ij}]$ be a GM of C.

Then by Lemma 7.2, the elements of $C^\perp$ are the vectors $v = (v_1, v_2, \ldots, v_n)$, since $v \in C^\perp \iff v \cdot G^T = 0$, satisfying

$$\sum_{j=1}^{m} g_{ij} v_j = 0, \ \forall \ i = [k]$$

$\longrightarrow$ this is a system of k independent homogeneous equations in n unknowns, with solution space $C^\perp$. From linear algebra we know that it has dimension $n-k$. To see it:

if codes C_1 and C_2 are equivalent, then $C_1^\perp$, $C_2^\perp$ are also equivalent. Hence is enough to show that $\dim(C^\perp) = n-k$, where C is a standard form GM:

$$C = \begin{pmatrix} 1 & \ldots & 0 & a_{1,1} & \ldots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \ldots & 1 & a_{k,1} & \ldots & a_{k,n-k} \end{pmatrix}$$

Then

$$C^\perp = \{(v_1, \ldots, v_n) \in V(n,q) \mid v_i + \sum_{j=1}^{n-k} a_{i,j} v_{k+j} = 0, \ i = 1, 2, \ldots, k\}$$

For each of the q^{n-k} choices of $(v_{k+1}, \ldots, v_n)$, $\ \exists! \ (v_1, v_2, \ldots, v_n) \in C^\perp$.

Hence $|C^\perp| = q^{n-k}$, therefore $dim(C^\perp) = n-k$.

□

Theorem 7.5. for any $[n,k]$-code C, $(C^\perp)^\perp = C$.

Proof. clearly $C \subseteq (C^\perp)^\perp$, since every vector in C is orthogonal to every vector in $C^\perp$.

But $dim((C^\perp)^\perp) = n - (n-k) = k = dim(C)$, therefore $C = (C^\perp)^\perp$. □

3.2 PCM (parity-check matrix)

Definition (PCM)**.** A *parity-check matrix* (PCM) H for a $[n,k]$-code C is a generator matrix of $C^\perp$.

$\Longrightarrow$ H is a $(n-k) \times n$ matrix satisfying $GH^T = 0$.

From 7.2 and 7.5,
if H a PCM of C, then

$$C = \{x \in V(n,q) \mid xH^T = 0\}$$

$\implies$ Any linear code is completely specified by a PCM.

Intuition . i. PCM: G specifies how to build a codeword, the PCM H specifies the rules a vector must follow to be considered a valid codeword.

ii. dual-code: orthogonal "shadow", see C as a plane through 3D origin; $C^\perp$ is the (normal vector) line emerging from the plane.

Rows of the PCM are the basis for the dual-code $C^\perp \implies$ ie. the rules that define C are themselves a code.

Theorem 7.6. if $G = [I_k|A]$ is the standard form generator matrix of an $[n,k]$-code C, then a PCM for C is $H = [-A^T \mid I_{n-k}]$.
eg.

$$G = \begin{bmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{bmatrix}$$

$$H = \begin{bmatrix} -a_{1,1} & \dots & -a_{1,n-k} & 1 & \dots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ -a_{k,1} & \dots & -a_{k,n-k} & 0 & \dots & 1 \end{bmatrix}$$

which, in binary, the minus signs are unnecessary.

Definition (Standard form)**.** a PCM that is $H = [B \mid I_{n-k}]$

3.3 Syndrome

Definition (Syndrome)**.** let H a PCM of a $[n,k]$-code C.
Then $\forall\ y \in V(n,q)$, the $1 \times (n-k)$ row vector

$$S(y) = yH^T$$

is called the *syndrome* of y.

Remark . i. for $h_1, h_2, \dots, h_{n-k}$ being rows of H, then

$$S(y) = (y \cdot h_1, y \cdot h_2, \dots, y \cdot h_{n-k})$$

ii. $S(y) = 0 \iff y \in C$

Intuition . Syndrome is the ”sympthom” of the error. It ignores the valid part of the message and isolates the noise.

$$S(y) = yH^T = (c+e)H^T = \underbrace{cH^T}_{0} + eH^T = 0 + eH^T = eH^T$$

Syndrome allows the receiver to identify what went wrong without knowing what was originally sent.

Lemma 7.8. two vectors u, v are the same ocset of C iff they have the same syndrome.

Proof. if u, v in the same coset of C

$$\begin{aligned}
&\implies u + C = v + C \\
&\implies u - v \in C \\
&\implies (u - v)H^T = 0 \\
&\implies uH^T = vH^T \\
&\implies S(u) = S(v)
\end{aligned}$$

□

Corollary 7.9. There is a one-to-one ccorrespondence between cosets and syndromes.

$\longrightarrow$ for small n, it's fast to locate the received vector y in the array, but for large n: use syndrome to find which coset (ie. which row of the array) contains y.

Example 7.10. (from example 6.5)

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

by 7.6,

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Hence the syndromes of the coset leaders are

$$\begin{aligned}
S(0000) &= 00 \\
S(1000) &= 11 \\
S(0100) &= 01 \\
S(0010) &= 10
\end{aligned}$$

The standard array becomes:

cosetleaders				*syndromes*
0000	1011	0101	1110	00
1000	0011	1101	0110	11
0100	1111	0001	1010	01
0010	1001	0111	1100	10

so, decoding algorithm: when a vector y is received, calculate $S(y) = yH^T$, locate $S(y)$ in the *syndromes* column of the array.

Locate y in the corresponding row and decode as the codeword at the top of the column containing y.

$\longrightarrow$ in practice: *syndrome lookup table*:

syndrome z	coset leader $f(z)$
00	0000
11	1000
01	0100
10	0010

decoding procedure:

1. receive y, calculate $S(y) = yH^T$
2. let $z = S(y)$, locate z in the first column of the looku table
3. decode y as $y - f(z)$

3.4 Incomplete decoding

For $d(C) = 2t + 1$ (or $2t + 2$), guarantee the correction of $\leq t$ errors in any codewordd and also detect some cases of more than t errors.

$\longrightarrow$ Arrange the cosets in order of increasing weight of the coset leaders.

Divide the array into a top part and a bottom part; top: cosets whose leaders have weights $\leq t$.

- If y in top part: decode as usual (assuming $\leq t$ errors).
- If y in bottom part: conclude that $> t$ errors have occurred, ask for retransmission.

3.5 Exercises

Exercise 7.1. prove Lemma 7.1

Proof. For any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

i. $u \cdot v = v \cdot u$

$$u \cdot v = \sum_{i=1}^{n} u_i \cdot v_i = \sum_{i=1}^{n} v_i \cdot u_i = v \cdot u$$

ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

$$\begin{aligned}
(\lambda u + \mu v)w &= \sum (\lambda u_i + \mu v_i) w_i \\
&= \sum (\lambda u_i w_i + \mu v_i w_i) \\
&= \sum (\lambda u_i w_i) + \sum (\mu v_i w_i) \\
&= \lambda \sum (u_i w_i) + \mu \sum (v_i w_i) \\
&= \lambda \cdot (u \cdot w) + \mu \cdot (v \cdot w)
\end{aligned}$$

□

Exercise 7.2. prove that if E_n is the binary even weight code of length n, then $E_n^{\perp}$ is the repetition code of length n.

Proof. from exercise 5.2:

$$E_n = \{x \in V(n,2) \mid w(x) \text{ is even}\} = \{(x_1, \ldots, x_n) \in F_2^n \mid x_1 + \ldots + x_n = 0\}$$

GM of E_n:

$$G = \left[\begin{array}{c|c} & 1 \\ I_{n-1} & \vdots \\ & 1 \end{array}\right]$$

thus, by 7.6, PCM is $H = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}$.
If H is a PCM for C, then the rows of H generate $C^{\perp}$.
Therefore

$$E_n^{\perp} = < \underbrace{[1\ 1\ \cdots\ 1]}_{generator} >$$

then it's codewords are:

$$\begin{aligned}
0 \cdot (1, \ldots, 1) = (0, \ldots, 0) \\
1 \cdot (1, \ldots, 1) = (1, \ldots, 1)
\end{aligned}$$

$$\implies E_n^{\perp} = \{00 \ldots 00,\ 11 \ldots 1\}$$

ie. the repetition code of length n. □

Exercise 7.4. for a binary linear code with PCM H, show that the transpose of the syndrome of a received vector is equal to the sums of those columns of H corresponding to where the errors occurred.

Proof. Let e the error, y the vector received, and x the original codeword. Then $y = x + e$.

Then

$$S(y) = yH^T = (x+e)H^T = \underbrace{xH^T}_{\substack{0,\text{since } H \\ \text{is PCM of } C \\ \text{and } x \in C}} + eH^T = eH^T$$

So,

$$S(y)^T = e^T H = \sum_{j=1}^{n} e_j \cdot H_j$$

where H_j is the j-th column of H. $\square$

Exercise 7.6. Let C be the ternary linear code with GM

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

(a) Find a generator matrix in standard form.

(b) Find a parity check matrix (PCM) for C in standard form.

(c) Use syndrome decoding to decode the received vectors: $2122, 1201, 2222$

Proof. (a)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

thus $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ is PCM in standard form.

(b) since $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ then

$$H = \begin{bmatrix} -2 & -2 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

(c)

$$S(y) = yH^T = S(y) = (y \cdot h_1,\ y \cdot h_2).$$

w/ h_i the rows of H.

compute the lookup table (syndrome table):

e	$S(e)$
0000	00
1000	11
2000	22
0100	12
0200	21
0010	10
0020	20
0001	01
0002	02

left: weight-1 errors, since we're over F_3; right: coset leaders for the 9 cosets.

i. $y = 2121$:

$$S(y) = \begin{bmatrix} 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = [5,5] \equiv [2,2].$$

$S(y) = [2,2]$, corresponds (in the table) to $e = 2000$.

$$\implies c = y - e = 2121 - 2000 = 0121$$

ii. $y = 1201$

$$S(y) = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} H^T = [3,6] = [0,0].$$

corresponds to $e = 0000$

$$\implies c = y - e = 1201$$

iii. $y = 2222$

$$S(y) = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix} H^T = [6,8] = [0,2].$$

corresponds to $e = 0002$

$$\implies c = y - e = 2222 - 0002 = 2220$$

□

4 Hamming Codes

Definition (Binary Hamming Code)**.** let r a positive integer, H a $r \times (2^r - 1)$ matrix whose columns are the distinct non-zero vectors of $V(r,2)$.

The code having H as its PCM is called a *binary Hamming code*, denoted $Ham(r,2)$.

Remark . $Ham(r,2)$ has length $n = 2^r - 1$ and dimension $k = n - r$.

Redundancy of the code: $r = n - k$, the number of check symbols in each codeword.

Exercise 8.1. i. $r = 2$: $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

by Theorem 7.6, $G = [1\ 1\ \underbrace{1}_{I}]$

so $Ham(2,2)$ is just the binary triple repetition code.

ii. $r = 3$: a PCM for $Ham(3, 2)$ is

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

(have taken columns in the natural order of increasing binary numbers from 1 to 7)

H in standard form: rearrange columns:

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

by Theorem 7.6, a generator matrix for $Ham(3, 2)$ is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Theorem 8.2. the binary Hamming code $Ham(r, 2)$, for $r \geq 2$,

1. is a $[2^r - 1, 2^r - 1 - r]$-code
2. has minimum distance 3 (hence is single-error-correcting)
3. is a *perfect code*

Lemma (Decoding with a binary Hamming code)**.** $Ham(r, 2)$ is a perfect single-error-correcting code $\Longrightarrow$ the coset leader are precisely the 2^r $(= n + 1)$ vectors of $V(n, 2)$ of weight ≤ 1.

Let j-th column of H be the binary representation of j. Decoding algorithm:

1. receive vector y, calculate its syndrome: $S(y) = yH^T$
2. if $S(y) = 0$, assume $y = c$ (original sent codeword)
3. if $S(y) \neq 0$, assuming a single error, $S(y)$ gives the binary representation of the error position, and the error can be corrected.

Example . $H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$, if $y = 1101011$ then

$$S(y) = y \cdot H^T = 110$$

$\Longrightarrow$ 6-th position, and so we decode y as 1101001

4.1 Extended binary Hamming codes

$\hat{Ham}(r,2)$, obtained from $Ham(r,2)$ by adding an overall parity-check.

Not ideal for complete decoding, but good for incomplete decoding (correct any single error and detect any double error).

Let H be a PCM for $Ham(r,2)$. A PCM $\hat{H}$ for $\hat{Ham}(r,2)$:

$$\hat{H} = \begin{bmatrix} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ 1 & \cdots & 1 & 1 \end{bmatrix}$$

If H is taken with columns in increasing order of binary numbers, we have a neat incomplete decoding algorithm:

Example 8.3. $r = 3$, $\hat{Ham}(3,2)$. Received y, $S(y) = y\hat{H}^T = (s_1, s_2, s_3, s_4)$

i. if $s_4 = 0$ and $(s_1, s_2, s_3) = 0$: assume no errors

ii. if $s_4 = 0$ and $(s_1, s_2, s_3) \neq 0$: assume ≥ 2 errors and ask retransmission

iii. if $s_4 = 1$ and $(s_1, s_2, s_3) = 0$: assume 1 error in the last position

iv. if $s_4 = 1$ and $(s_1, s_2, s_3) \neq 0$: assume 1 error in the j-th position, where j is the number binary-represented by (s_1, s_2, s_3)

Theorem 8.4. suppose C a linear $[n,k]$-code over $GF(q)$ with PCM H. Minimum distance of C is d iff any $d-1$ columns of H are linearly independent but some d columns are linearly dependent.

Proof. by 5.2, min. distance of C (ie. $d(C)$) = smallest of the weights of the non-zero codewords.

Let $x = x_1x_2\ldots x_n \in V(n,q)$, then

$$\begin{aligned} x \in C &\iff xH^T = 0 \\ &\iff x_1H_1 + x_2H_2 + \ldots + x_nH_n = 0 \end{aligned}$$

where H_i denote the columns of H.

$\implies$ each codeeword x of weight d corresponds with a set of d linearly dependent columns of H.

On the other hand, if $\exists$ a set of $d-1$ lin.dependent columns of H, say $H_{i_1}, H_{i_2}, \ldots, H_{i_{d-1}}$, then $\exists$ $x_{i_1}, \ldots, x_{i_{d-1}}$ scalars, not all zero, such that

$$x_{i_1}H_{i_1} + x_{i_2}H_{i_2} + \ldots + x_{i_{d-1}}H_{i_{d-1}} = 0$$

But then, the vector

$$x = (0 \dots 0 x_{i_1} 0 \dots x_{i_2} 0 \dots 0 x_{i_{d-1}} 0 \dots 0)$$

having x_{i_j} in the j-th position for $j \in [d-1]$, would satisfy $xH^T = 0$, and so would be a non-zero codeword of weight $< d$. □

4.2 q-ary Hamming codes

For C a inear code with $d(C) = 3$ (min.dist), any 2 columns of a PCM H must be linearly independent

$\Longrightarrow$ columns of H must be non-zero, and no column can be a scalar multiple of another

Any $0 \neq v \in V(r, q)$ vector has $q - 1$ non-zero scalar multiples, forming the set

$$\{\lambda v \mid \lambda \in GF(q), \lambda \neq 0\}$$

$\longrightarrow$ the $q^r - 1$ non-zero vectors of $V(r, q)$ may be partitioned into $\frac{q^r-1}{q-1}$ such sets, called *classes.*

$\longrightarrow$ two vectors are scalar multiples of each other if they are in the same class

By choosing one vector from each class, obtain a set of $\frac{q^r-1}{q-1}$ vectors, where not two are linearly dependent.

By Theorem 8.4, taking these as columns of h gives a PCM for a

$$\left[\frac{q^r - 1}{q - 1}, \frac{q^r - 1}{(q-1) - r}, 3\right]\text{-code}$$

$\Longrightarrow$ called *q-ary Hamming code*, $Ham(r, q)$.

$\longrightarrow$ PCM is unique up to permutation of columns

$\Longrightarrow$ Hamming codes are linear codes

Theorem 8.6. *$Ham(r, q)$ is a perfect single-error-correcting code.*

Proof. $Ham(r, q)$ is constructed to be a $(n, M, 3)$-code, with $n = \frac{q^r-1}{q-1}$ and $M = q^{n-r}$.

With $t = 1$, the LHS of the sphere-packing bound becomes

$$q^{n-r} \cdot (1 + n(q-1)) = q^{n-r} \cdot (1 + q^r - 1) = q^n$$

which is the RHS, so $Ham(r, q)$ is a perfect code. □

Decoding q-ary Hamming code Since Ham is perfect single-error correcting code, the coset leaders, other than 0, are the vectors of weight 1.

Syndrome of such coset leader:

$$S(0 \dots 0 \; b \; 0 \dots 0) = (0 \dots 0 \; b \; 0 \dots 0) \cdot H^T = b \cdot H_j^T$$

where H_j is the j-th column of H, and b is at the j-th position of y.

Decode:

- given received y, calculate $S(y) = yH^T$
- if $S(y) = 0$, assume no errors
- if $S(y) \neq 0$, then $S(y) = b \cdot H_j^T$ for some b and j
  correct by: subtract b from the j-th entry of y.

Example:

$$q = 5,\ H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 3 & 4 \end{bmatrix},\ y = 203031$$

Then,

$$\begin{aligned} S(y) = y \cdot H^T &= 2 \cdot H_1^T + 3 \cdot H_3^T + 3 \cdot H_5^T + 1 \cdot H_6^T \\ &= 2 \cdot [0\ 1] + 3 \cdot [1\ 1] + 3 \cdot [1\ 3] + 1 \cdot [1\ 4] \\ &= [0\ 2] + [3\ 3] + [3\ 9] + [1\ 4] \\ &= [7\ 8] = [2\ 3] = \underbrace{2}_{b} \cdot \underbrace{[1\ 4]}_{H_6^T} \pmod 5 \end{aligned}$$

$\implies$ error is in position 6-th, with magnitude $b = 2$
$\implies$ subtract b from the j-th entry of y: $1 - 2 \pmod 5 = -1 \pmod 5 = 4$, therefore $y' = 203034$.

5 MLCT - Main Linear Coding Theory problem

A good (n, M, d)-code has

- small n, for fast transmission of messages
- large M, to enable transmission of a wide variety of messages
- large d, to correct many errors

Denote by $A_q(n, d)$ the largest M such that $\exists$ a q-ary (n, M, d)-code.

Theorem 2.7. Suppose d is odd. Then a binary (n, M, d)-code exists iff a binary $(n + 1, M, d + 1)$-code exists.

Proof. □

Corollary 2.8. If d is odd, then $A_2(n + 1, d + 1) = A_2(n, d)$.
Equivalently, if d is even, then $A_2(n, d) = A_2(n - 1, d - 1)$.

Definition (Sphere)**.** $\forall\ u \in F_q^n$ (vector) and any integer $r \geq 0$, the *sphere* of radius r and centre u, denoted $S(u, r)$, is the set

$$\{v \in F_q^n \mid d(u, v) \leq r\}$$

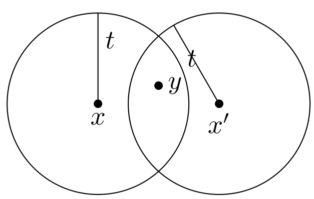
Theorem 1.9 & 2.12. a code C can

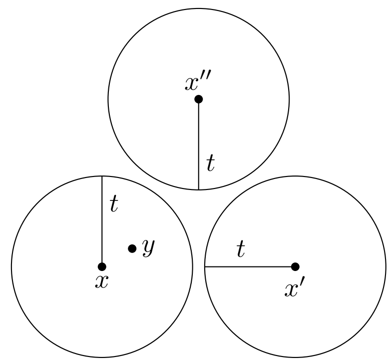
i. detect up to s errors in any codeword if $d(C) \geq s+1$

ii. correct up to t errors in any codeword if $d(C) \geq 2t+1$

This is, the spheres of radius t centered on the codewords of C are disjoint (ie. have no overlap).

So, if $\leq t$ errors occur, then the received vector y may be different from the sent codeword x, hence being different from the centre of the sphere $S(x,t)$, but can not 'escape' from the sphere, so is 'drawn back' to x by nearest neighbour decoding.





Definition (Perfect codes)**.** A code which achieves the sphere-packing bound is called a *perfect code*. Thus, for a perfect t-error-correcting code, the M spheres of radius t centred on codewords 'fill' the whole space F_q^n without overlapping.

In other words, every vector in F_q^n is at distance $\leq t$ from exactly one codeword.

Theorem (MLCT problem (Version 1))**.** For given length n, find the maximum dimension k such that $\exists$ an $[n,k,d]$-code over $GF(q)$. (Then, for this $k, B_q(n,d) = q^k$).

Remark . *redundancy* r of an $[n,k,d]$-code is just $r = n-k$ (the number of check symbols in a codeword).

Theorem (MLCT problem (Version 2))**.** For given redundancy r, find the maximum length n such that $\exists$ an $[n, n-r, d]$-code over $GF(q)$.

Definition ((n,s)-set)**.** a (n,s)-set in $V(r,q)$ is a set of n vectors in $V(r,q)$ with the property that any s of them are linearly independent.

Denote $max_s(r,q)$ the largest value of n for which $\exists$ a (n,s)-set in $V(r,q)$. ie.

- (n,s)-set: n vectors with s of them lin. independent

- $max_s(r,q)$: max n for which $\exists$ (n,s)-set

Definition (Packing problem)**.** The *packing problem* for $V(r,q)$ is that of determining the values of $max_s(r,q)$ and the optimal (n,s)-sets.

Theorem 14.1. There exists an $[n, n-r, d]$-code over $GF(q)$ iff $\exists$ an $(n, d-1)$-set in $V(r,q)$.

Proof. let C a $[n, n-r, d]$-code over $GF(q)$ with PCM H. Then by Theorem 8.4, the columns of H form a $(n, d-1)$-set in $V(r,q)$.

On the other hand, let K a $(n, d-1)$-set in $V(r,q)$. If we form a $r \times n$ matrix H with the vectors of K as its columns, by Theorem 8.4, H is the PCM of a $[n, n-r]$-code whose minimum distance is at least d. $\square$

Corollary 14.2. for given values q, d, r, the largest value of n for which there exists a $[n, n-r, d]$-code over $GF(q)$ is $max_{d-1}(r,q)$.

$\Longrightarrow$ the MLCT problem (V2) is the same as the packing problem of finding $max_{d-1}(r,q)$.

5.1 The projective geometry PG(r-1,q)

Let the vector space

$$V(r,q) = \{(a_1, a_2, \ldots, a_r) \mid a_i \in GF(q)\}$$

Definition (Projective Geometry)**.** $PG(r-1,q)$

- *points* are the one-dimensional subspaces of $V(r,q)$
- *lines* are the two-dimensional subspaces of $V(r,q)$

Each point $P \in PG(r-1,q)$, as a subspace of $V(r,q)$ of dimension 1, is generated by a single non-zero vector.

So if $a = (a_1, a_2, \ldots, a_r) \in P$, then $P = \{\lambda a \mid \lambda \in GF(q)\}$.

$\Longrightarrow$ for $a = (a_1, a_2, \ldots, a_r)$, $b = (b_1, b_2, \ldots, b_r) \in P$, then

$$a = b \text{ in } PG(r-1,q) \text{ iff } a = \lambda b \text{ in } V(r,q)$$

for some non-zero scalar λ.

Lemma 14.6. in $PG(r-1,q)$,

1. number of points is $\frac{q^r-1}{q-1}$
2. any two points lie on exactly one line
3. each line contains exactly $q+1$ points
4. each point lies on $\frac{q^{r-1}-1}{q-1}$ lines

Proof. 1. number of points is $\frac{q^r-1}{q-1}$:
since each of the $q^r - 1$ nonzero vectors in $V(r, q)$ has $q - 1$ nonzero scalar multiples, the number of points of $PG(r - 1, q)$ is $\frac{q^r-1}{q-1}$

2. any two points lie on exactly one line:
if a, b distinct points of $PG(r - 1, q)$, then the unique line through them consists of the points $\lambda a + \mu b$, where λ, μ are scalars not both zero.

3. each line contains exactly $q + 1$ points:
in (ii) there are $q^2 - 1$ choices for the pair (λ, μ), but we're identifying scalar multiples, so the number of distinct points on the line is

$$\frac{q^2 - 1}{q - 1} = q + 1$$

4. each point lies on $\frac{q^{r-1}-1}{q-1}$ lines:
let t be the number of lines on which a point P lies. Let the set

$$X = \{(Q, L) \mid Q \text{ is a point } \neq P,\ L \text{ is a line containing both } P \text{ and } Q\}$$

Count the members of X in two ways:
for each of the $\frac{q^r-1}{q-1} - 1$ choices of Q, $\exists!$ line containing P and Q. Thus,

$$|X| = \frac{q^r - 1}{q - 1} - 1 = \frac{q^r - q}{q - 1}$$

On the other hand, for each of the t lines through P, there are (by (iii)) q points Q other than P lying on L.

Thus, $|X| = tq$,

$$\implies tq = \frac{q^r - q}{q - 1}, \quad \implies t = \frac{q^{r-1} - 1}{q - 1}$$

□

Definition (Projective plane)**.** The $PG(2, q)$ is called the *projective plane* over $GF(q)$.

6 MDS Codes

Theorem 15.1. a $[n, n - r, d]$-code satisfies $d \leq r + 1$.

Proof. let C a $[n, n - r, d]$-code, let $G = [I_{n-r} \mid A]$ be a standard form GM of C.

Since A has r columns, the codewords whicch are rows of G have weight $\leq r + 1$, ie. $w(C) \leq r + 1$.

The result follows from Theorem 5.2,

$$\begin{aligned} &\Longrightarrow\ w(C) = d(C) \\ &\Longrightarrow\ d(C) \leq r+1. \end{aligned}$$

$\square$

Definition (MDS)**.** a $[n, n-r, r+1]$-code,
ie. a linear cocde of redundancy r whose min. distance is $r+1$,
is called a *maximum distance separable code*, MDS.

$\longrightarrow$ by Theorem 14.1, max length of a MDS code over $GF(q)$ is $max_r(r,q)$, the largest size of a (n,r)-set in $V(r,q)$.

Conjecture 15.2. if $2 \leq r \leq q$, then $max_r(r,q) = q+1$
(except $max_3(3,q) = max_{q-1}(q-1,q) = q+2$ if $q = 2^h$)

Theorem 15.3. if $r \geq q$, then $max_r(r,q) = r+1$. Any MDS code of redundancy $r \geq q$ is equivalent to a repetition code of length $r+1$.

Proof. repetition code of length $r+1$: a $[r+1, 1, r+1]$-code with GM $[1\ 1\ \cdots\ 1]$.
Hence $max_r(r,q) \geq r+1$.
Also, any $[r+1,1,r+1]$-code is equivalent to a repetition code.
Suppose $r \geq q$ and by contradiction that $max_r(r,q) \geq r+2$.
Then, $\exists$ a $[r+2, 2, r+1]$-code C over $GF(q)$.
$\longrightarrow$ C must be equivalent to a code having GM:

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & \dots & 1 \\ 0 & 1 & a_1 & a_2 & \dots & a_r \end{bmatrix}$$

in order that any linear combination of the rows of G has weight at least $r+1$ with $0 \neq a_i \in GF(q)$.
This implies $r \leq q-1$, contradiction. $\square$

Remark . the only binary MDS codes are
$V(n,2)$, the even weighted codes E_n, and repetition codes.

Theorem 15.4. let $2 \leq r \leq q$, let $a_1, a_2, \dots, a_{q-1} \in GF(q)$ nonzero.
Then,

$$H = \begin{bmatrix} 1 & 1 & \dots & 1 & 1 & 0 \\ a_1 & a_2 & \dots & a_{q-1} & 1 & 0 \\ a_1^2 & a_2^2 & \dots & a_{q-1}^2 & 1 & 0 \\ & & & & \vdots & \vdots \\ \vdots & \vdots & & \vdots & 0 & 0 \\ a_1^{r-1} & a_2^{r-1} & \dots & a_{q-1}^{r-1} & 0 & 1 \end{bmatrix}$$

is the PCM of a MDS $[q+1, q+1-r, r+1]$-code.
Equivalently, the columns of H form a $(q+1)$-arc in $PG(r-1,q)$.

Proof. the determinant of a matrix formed by any r columns of H is equal to the determinant of a Vandermonde matrix, and so is non-zero (see a proof of this at section 6.1). Thus any r columns of H are linearly independent. $\square$

Corollary 15.7. The dual code of a MDS code is also MDS.

Corollary 15.8. $\exists$ a MDS $[n, k]$-code over $GF(q)$ iff $\exists$ a MDS $[n, n-k]$-code over $GF(q)$.

6.1 Vandermonde determinant

This section contains a simple proof of the Vandermonde matrix determinant. The proof was found in [3], here it is written with small modifications and expanded. A different proof can be found in [4].

Lemma (Vandermonde determinant)**.**

$$\det\ V(x_1, ..., x_n) = \det \begin{pmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \dots & x_3^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{pmatrix} = \prod_{j<i}(x_i - x_j)$$

Proof. The equation is true for $n = 2$, as we have:

$$\det\ V(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1$$

We now assume that $n \geq 3$. Starting with column n and ending at column 2, we successively apply the following operation: subtract x_1 times column $i-1$ from column i. The determinant remains the same and the resulting matrix is:

$$= \det \begin{pmatrix} 1 & x_1 - (x_1 \cdot 1) & x_1^2 - (x_1 \cdot x_1) & \dots & x_1^{n-1} - (x_1 \cdot x_1^{n-2}) \\ 1 & x_2 - (x_1 \cdot 1) & x_2^2 - (x_1 \cdot x_2) & \dots & x_2^{n-1} - (x_1 \cdot x_2^{n-2}) \\ 1 & x_3 - (x_1 \cdot 1) & x_3^2 - (x_1 \cdot x_3) & \dots & x_3^{n-1} - (x_1 \cdot x_3^{n-2}) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n - (x_1 \cdot 1) & x_n^2 - (x_1 \cdot x_n) & \dots & x_n^{n-1} - (x_1 \cdot x_n^{n-2}) \end{pmatrix}$$

$$= \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & (x_2 - x_1) & x_2(x_2 - x_1) & \dots & x_2^{n-2}(x_2 - x_1) \\ 1 & (x_3 - x_1) & x_3(x_3 - x_1) & \dots & x_3^{n-2}(x_3 - x_1) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & (x_n - x_1) & x_n(x_n - x_1) & \dots & x_n^{n-2}(x_n - x_1) \end{pmatrix}$$

Applying the Laplace expansion along the first row and factoring out the scalar multiples $(x_i - x_1)$ in each column, we obtain:

$$= \det \begin{pmatrix} x_2 - x_1 & x_2^2 - x_1 x_2 & x_2^3 - x_1 x_2^2 & \dots & x_2^{n-1} - x_1 x_2^{n-2} \\ x_3 - x_1 & x_3^2 - x_1 x_3 & x_3^3 - x_1 x_3^2 & \dots & x_3^{n-1} - x_1 x_3^{n-2} \\ \vdots & \vdots & \vdots & & \vdots \\ x_n - x_1 & x_n^2 - x_1 x_n & x_n^3 - x_1 x_n^2 & \dots & x_n^{n-1} - x_1 x_n^{n-2} \end{pmatrix}$$

in each row, factor out $(x_i - x_1)$

$$= \det \begin{pmatrix} 1(x_2 - x_1) & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \dots & x_2^{n-2}(x_2 - x_1) \\ 1(x_3 - x_1) & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \dots & x_3^{n-2}(x_3 - x_1) \\ \vdots & \vdots & \vdots & & \vdots \\ 1(x_n - x_1) & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \dots & x_n^{n-2}(x_n - x_1) \end{pmatrix}$$

$$= \prod_{2 \leq i \leq n} (x_i - x_1) \quad \det \begin{pmatrix} 1 & x_2 & x_2^2 & \dots & x_2^{n-2} \\ 1 & x_3 & x_3^2 & \dots & x_3^{n-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-2} \end{pmatrix}$$

now, by induction

$$= \prod_{i=2}^{n} (x_i - x_1) \quad \cdot \prod_{2 \leq j < i \leq n} (x_i - x_j) \quad = \prod_{1 \leq j < i \leq n} (x_i - x_j)$$

□

7 Cyclic Codes

This section contains notes both from Hill [1] and Lint [2] books. Hill's notes are enumerated as $X.Y$ while Lint's notes as $X.Y.Z$.

7.1 Recap on ideals

More about ideals at the *Commutative Algebra notes* [5].

In the cyclic codes context, we're interested in ideals because

- cyclic codes are ideals of $R_n = \mathbb{F}_q[x]/(x^n - 1)$.
- R_n is a PID: every ideal in R_n is principal.

Definition (ideal)**.** $I \subset R$ (R ring) such that $0 \in I$ and $\forall x \in I,\ r \in R,\ xr, rx \in I$.

ie. I absorbs products in R.

Definition (prime ideal)**.** if $a, b \in R$ with $ab \in P$ and $P \neq R$ (P a prime ideal), implies $a \in P$ or $b \in P$.

Definition (principal ideal)**.** generated by a single element, (a).
(a): principal ideal, the set of all multiples xa with $x \in R$.

Definition (prime spectrum - $Spec(A)$)**.** set of prime ideals of A. ie.

$$Spec(A) = \{P \mid P \subset A \text{ is a prime ideal}\}$$

Definition (integral domain)**.** Ring in which the product of any two nonzero elements is nonzero.
ie. no zerodivisors.
ie. $\forall\ 0 \neq a,\ 0 \neq b \in A,\ ab \neq 0 \in A$.
Every field is an integral domain, not the converse.

Definition (principal ideal domain - PID)**.** integral domain in which every ideal is principal. ie. ie. $\forall I \subset R,\ \exists\ a \in I$ such that $I = (a) = \{ra \mid r \in R\}$.

7.2 Cyclic Codes intro

Definition (cyclic code)**.** a code C is cyclic if

i. C is a linear code

ii. any cyclic shift of a codeword is also a codeword
ie. when $a_0 a_1 \dots a_{n-1} \in C$,
then so $a_{n-1} a_0 a_1 \dots a_{n-2} \in C$.

Remark (Notation)**.** From now on,

$$F_q = \mathbb{F}_q = GF(q)$$

$F_q[x]$ is a ring; and for $f(x)$ irreducible in $F_q[x]$, then $\frac{F_q[x]}{f(x)}$ is a field.
From now on,

$$f(x) = x^n - 1 \quad \Longrightarrow \quad F_q[x]/(x^n - 1) = R_n$$

Observe:
since $x^n \cong 1 \bmod x^n - 1$, to reduce any polynomial by modulo $x^n - 1$ can just replace x^n by 1, x^{n+1} by x, x^{n+2} by x^2, etc.

Correspondence:

$$a_0a_1\dots a_{n-1} \in V(n,q) \iff a(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in R_n$$

In R_n,

$$\begin{aligned} x \cdot a(x) &= a_0 \cdot x + a_1x^2 + \dots + \underbrace{a_{n-1}x^n}_{=a_{n-1}\cdot 1} \\ &= a_{n-1} + a_0x + \dots + a_{n-2} \cdot x^{n-1} \in R_n \end{aligned}$$

ie. $a_{n-1}a_0a_1\dots a_{n-2} \in V(n,q)$

In R_n:

→ multiply by x corresponds to performing a single cyclic shift

→ multiply by x^m corresponds to a cyclcic shift through m positions

Theorem 12.6 (Algebraic characterization of cyclic codes)**.** a code $C \in R_n$ is a cyclic code iff

i. $a(x), b(x) \in C \implies a(x) + b(x) \in C$

ii. $a(x) \in C$ and $r(x) \in R_n \implies r(x)a(x) \in C$
ie. C closed under multiplication by any element of R_n

Proof. i. let $C \in R_n$ cyclic code, then C is linear, so (i) holds.

ii. let $a(x) \in C$ and $r(x) = r_0 + r_1x + \dots + r_{n-1}x^{n-1} \in R_n$.

Since mult. by x corresponds to a cyclic shift, then $x \cdot a(x) \in C$, thus

$$x \cdot (x \cdot a(x)) = x^2 \cdot a(x) \in C$$

and so on.

Hence,

$$r(x) \cdot a(x) = r_0 \cdot a(x) + r_1 \cdot x \cdot a(x) + \dots + r_{n-1} \cdot x^{n-1} \cdot a(x) \in C$$

since each summand is in C; thus (ii) holds.

Conversely, suppose (i) and (ii) hold;
taking $r(x)$ to be a scalar, the conditions imply that C is linear.
Taking $r(x) = x$, in (ii) shows that C is cyclic.

□

Remark . i. Cyclic codes are the ideals of the ring R_n.

ii. Every ideal in R_n is a *principal ideal*.
 ie. R_n is a PID.

Theorem 6.1.3. a linear code C in $\mathbb{F}_q^n$ is cyclic iff C is an ideal in $\mathbb{F}_q[x]/(x^n - 1)$.

Proof. If C ideal in $\mathbb{F}[x]/(x^n - 1)$ and $c(x) = c_0 + c_1 x + \ldots + c_{n-1}x^{n-1} \in C$, then

$$\begin{aligned} x \cdot c(x) &= c_0 x + c_1 x \cdot x + \ldots + \underbrace{c_{n-1}x^{n-1} \cdot x}_{=c_{n-1}\cdot 1 \pmod{x^n - 1}} \\ &= c_{n-1} + c_0 x + c_1 x^2 + \ldots + c_{n-2}x^{n-1} \in C \\ &\implies (c_{n-1}, c_0, c_1, \ldots, c_{n-2}) \in C \end{aligned}$$

Conversely, if C cyclic, then $\forall c(x) \in C,\ x \cdot c(x) \in C$.
Therefore $x^i \cdot c(x) \in C\ \forall i \in [n]$
and since C is linear, $a(x) \cdot c(x) \in C$
ie.

$$a_0 \cdot c(x) + a_1 \cdot x \cdot c(x) + a_2 \cdot x^2 \cdot c(x) + \ldots + a_{n-1} \cdot x^{n-1} \cdot c(x) \in C$$

$\forall a(x) \in \mathbb{F}_q[x]/(x^n - 1)$. □

Corollary . $\mathbb{F}_q[x]/(x^n - 1)$ is a PID (Principal Ideal Domain).
Thus C consists of the multiples of $g(x) \in C$ monic and of lowest degree.
Therefore,

$\implies$ $g(x)$ is the *generator polynomial* (more at 12.9) of C: $C = \langle g(x) \rangle$.

$\implies$ $g(x)$ is divisor of $x^n - 1$, ie. $g(x)|(x^n - 1)$.
since otherwise $gcd(x^n - 1, g(x))$ would be a polynomial in C with degree $< deg(g(x))$, a contradiction.

Let $f(x) \in R_n$, $\langle f(x) \rangle$ the subset of R_n consisting of all multiples of $f(x)$ (reduced modulo $x^n - 1$). ie.

$$\langle f(x) \rangle = \{ r(x) \cdot f(x) \mid r(x) \in R_n \}$$

Theorem 12.7. for any $f(x) \in R_n$, the set $\langle f(x) \rangle$ is a cyclic code; named the *code generated by* $f(x)$.

Proof. check conditions (i), (ii) of Theorem 12.6:

i. if $a(x)f(x),\ b(x)f(x) \in \langle f(x) \rangle$ then

$$a(x)f(x) + b(x)f(x) = (a(x) + b(x))f(x) \in \langle f(x) \rangle$$

ii. if $a(x)f(x) \in \langle f(x) \rangle$ and $r(x) \in R_n$, then

$$r(x)(a(x)f(x)) = (r(x) \cdot a(x))f(x) \in \langle f(x) \rangle$$

□

Example 12.8. $C = \langle 1 + x^2 \rangle \in R_3$, with $GF(2)$.
R_3 has $2^3 = 8$ elements.
Multiply $1 + x^2$ by each of the 8 elements of R_3 (reducing modulo $x^3 - 1$), obtaining 4 distinct codewords:

$$0, 1 + x, 1 + x^2, x + x^2$$

Thus C is the code $\{000, 110, 101, 011\}$.

7.3 Generator polynomial and check polynomial

Theorem 12.9. let C a nonzero cyclic coded in R_n. Then

i. $\exists!\ g(x)$ monic, of smallest degree in C

ii. $C = \langle g(x) \rangle$

iii. $g(x)$ is a factor of $x^n - 1$

Proof. i. $\exists!\ g(x)$ monic, of smallest degree in C:
let $g(x), h(x) \in C$ monic and of smallest degree. Then $g(x) - h(x) \in C$ with smaller degree, which is a contradiction if $g(x) \neq h(x)$

ii. $C =< g(x) >$:
let $a(x) \in C$. By division algorithm for $F_q[x]$,

$$a(x) = q(x)g(x) + r(x)$$

with $deg(r(x)) < deg(g(x))$.

But $r(x) = a(x) - q(x)g(x) \in C$ by the properties of cyclic codes (Theorem 12.6, $a, g \in C$ and C an ideal)

By minimality of $deg(g(x))$, must have $r(x) = 0$, therefore

$$a(x) \in \langle g(x) \rangle \forall a(x) \in C$$

$$\implies C = \langle g(x) \rangle$$

iii. $g(x)$ is a factor of $x^n - 1$:
by division algorithm,

$$x^n - 1 = q(x)g(x) + r(x)$$

with $deg(r(x)) < deg(g(x))$.

But then $r(x) = -q(x)g(x) \bmod x^n - 1$, ie. $r(x)$ is a multiple of $g(x)$, and so $r(x) \in \langle g(X) \rangle$.

By the minimality of $deg(g(x))$, we must have $r(X) = 0$, which implies

$$x^n - 1 = q(x) \cdot g(x) + 0$$

$\implies g(x)$ is a factor of $(x^n - 1)$.

$\square$

Definition (Generator polynomial)**.** let C cyclic code, monic polynomial of least ddegree (given by Theorem 12.9) is called the *generator polynomial* of C.

Remark C may contain other polynomials than the generator polynomial which also generate C.

Example 12.10. All cyclic codes of length 3:
ie. ideals in $\mathbb{F}_2[x]/(x^3-1) \implies$ factor x^3-1 in $\mathbb{F}_2$, every cyclic code corresponds to a divisor of $x^3 - 1$.

$$x^3 - 1 = (x+1) \cdot (x^2 + x + 1)$$

so by Theorem 12.9(ii):

generator poly	code in R_3	Corresponding code in $V(3,2)$
1	all of R_3	all of $V(3,2)$
$x+1$	$\{0, 1+x, x+x^2, 1+x^2\}$	$\{000, 110, 011, 101\}$
x^2+x+1	$\{0, 1+x+x^2\}$	$\{000, 111\}$
x^3-1	$\{0\}$	$\{000\}$

Notice that the 2nd column contains the multiples $\pmod{x^3 - 1}$ of the polynomials at the 1st column.

Lemma 12.11. let $g(x) = g_0 + g_1 + \ldots + g_r x^r$ be the generator polynomial. Then g_0 is nonzero.

Proof. by contradiction, let $g_0 = 0$. Then $x^{n-1} \cdot g(x) = x^{-1} \cdot g(x) \in C$ with degree $r - 1$.

This contradicts minimality of $deg(g(x))$; thus $g_0 \neq 0$. $\square$

Theorem 12.12. let C cyclic code with generator polynomial $g(x) = g_0 + g_1 x + \ldots + g_r x^r$. Then $dim(C) = n - r$, and a GM for C is

$$G = \begin{pmatrix} g_0 & g_1 & g_2 & \cdots & g_r & 0 & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & g_2 & \cdots & g_r & 0 & \cdots & 0 \\ 0 & 0 & g_0 & g_1 & g_2 & \cdots & g_r & \cdots & 0 \\ & \vdots & & & & & & \vdots & \\ 0 & 0 & \cdots & 0 & g_0 & g_1 & g_2 & \cdots & g_r \end{pmatrix}$$

Proof. the $n-r$ rows of G are linear independent because the echelon of nonzero g_0's with 0's below.

These $n-r$ represent the codewords $g(x), x \cdot g(x), x^2 g(x), \ldots, x^{n-r-1} g(x)$.

It remains to show that every codeword in C can be expressed as a linear combination of them.

From Theorem 12.9.(ii), if $a(x) \in C$ then $a(x) = q(x) \cdot g(x)$ for $q(x) \in \mathbb{F}_q[x]$.

Since $deg(a(x)) < n, \quad deg(q(x)) < n - r$.

Hence

$$\begin{aligned} q(x)g(x) &= (q_0 + q_1 x + \ldots + q_{n-r-1} x^{n-r-q}) \cdot g(x) \\ &= q_0 g(x) + q_1 x g(x) + \ldots + q_{n-r-1} x^{n-r-q} g(x) \end{aligned}$$

which is the desired linear combination. □

Definition (check polynomial)**.** For generator polynomial $g(x)$, which we know is a factor of $x^n - 1$.

Then, the *check polynomial* $h(x)$ is such that

$$x^n - 1 = g(x)h(x)$$

Since $g(x)$ is monic, so also is $h(x)$. By Theorem 12.12,

$$deg(g(x)) = n - k \implies deg(h(x)) = k$$

Theorem 12.14. $c(x) \in R_n$ is $c(x) \in C$ iff $c(x) \cdot h(x) = 0$.

Proof. in $R_n: \; g(x) \cdot h(x) = x^n - 1 = 0 \bmod x_n - 1$,
hence

$$c(x) \in C \overset{implies}{\implies} c(x) = a(x) \cdot g(x)$$

for some $a(x) \in R_n$.

$$\begin{aligned} &\implies c(x) \cdot h(x) = \overbrace{a(x) \cdot g(x)}^{c(x)} \cdot h(x) \\ &\text{(note that } g(x) \cdot h(x) = 0) \\ &\qquad\qquad\qquad = a(x) \cdot 0 = 0 \end{aligned}$$

$\implies c(x) \cdot h(x) = 0$ as stated.

Conversely, suppose $c(x) \cdot h(x) = 0$.

By division algorithm, $c(x) = q(x) \cdot g(x) + r(x)$ with $deg(r) < n - k$.

Then, $c(x) \cdot h(x) = 0$ implies

$$\begin{aligned} (q(x) \cdot g(x) + r(x)) \cdot h(x) &= 0 \\ q(x) \cdot g(x) \cdot h(x) + r(x) \cdot h(x) &= 0 \\ \text{recall } g(x) \cdot h(x) = x^n - 1, \text{ thus,} & \\ 0 + r(x) \cdot h(x) &= 0 \\ r(x) \cdot h(x) &= 0 \\ \text{ie.} r(x) \cdot h(x) = 0 \quad (\text{mod } x^n - 1) & \end{aligned}$$

But $deg(r(x) \cdot h(x)) < n - k + k = n$, so $r(x) \cdot h(x) = 0$ in $\mathbb{F}_q$.
Hence $r(x) = 0$, thus

$$\begin{aligned} c(x) &= q(x) \cdot g(x) \in C \\ &\implies c(x) \in C \end{aligned}$$

since C is an ideal of R_n and $g(x) \in C$ and $q(x) \in R_n$. □

7.4 Maximal and minimal cyclic codes

Let $x^n - 1 = f_1(x) \cdot f_2(x) \cdot \ldots \cdot f_t(x)$, with $f_i(x)$ irreducible polynomials.

Definition 6.1.6 (maximal and minimal cyclic codes)**.** the cyclic code generated by $f_i(x)$ is called a *maximal cyclic code* (since it's a maximal ideal), denoted M_i^+.

The code generated by $(x^n-1)/f_i(x)$ is called a *minimal cyclic code*, denoted M_i^-. Also named *irreducible code*.

Definition (Zeros of a cyclic code)**.** Let $x^n - 1 = f_1(x) \cdot \ldots \cdot f_t(x)$. Let β_i be a zero of $f_i(x)$ in some extension field of $\mathbb{F}_q$
ie. $f_i(\beta) = 0$, and $f_i(x)$ is the minimal polynomial of β_i.

$\implies M_i^+$ is the set of polynomials $c(x)$ for which $c(\beta_i) = 0$.

In general, a cyclic code can be specified by requiring that all codewords have certain zeros: require that all codewords have β_i points as zeros. For example:

Start by any set $\alpha_1, \alpha_2, \ldots, \alpha_s$. Define the code C by

$$c(x) \in C \text{ iff } c(\alpha_i) = 0 \ \forall i = 1, 2, \ldots, s$$

then C is cyclic, and the generator polynomial is the least common multiple (lcm) of the minimal polynomials of $\alpha_1, \alpha_2, \ldots, \alpha_s$.

7.5 Hamming cyclic codes

Theorem 6.3.1. let $n := (q^m - 1)/(q - 1)$, let β a primitive n-th root of unity in $\mathbb{F}_{q^m}$. Let $(m, q - 1) = 1$.

The cyclic code $C := \{c(x)|c(\beta) = 0\}$ is equivalent to the $[, n-m]$ Hamming code over $\mathbb{F}_q$.

Proof. since $n = (q - 1) \cdot (q^{m-2} + 2q^{m-3} + \ldots + m - 1) + m$ have $(n, q - 1) = (m, q - 1) = 1$.

Therefore $\beta^{i(q-1)} \neq 1 \ \forall i \in [n]$, ie. $\beta^i \neq \mathbb{F}_q \ \forall i \in [n]$.

Columns of H, which are the representations of $1, \beta, \beta^2, \ldots, \beta^{n-1}$ as vectors in $\mathbb{F}_{q^m}$, are pairwise linearly independent over $\mathbb{F}_q$.

So H is the PCM of an $[n, n - m]$ Hamming code. □

Example 6.3.2. smallest extension of $\mathbb{F}_2$ which contains a primitive 9-th root of unity is $\mathbb{F}_{2^6}$.

Let α a primitive element of the field, then $\alpha^6 3 = 1$,and $\beta := \alpha^7$ is a primitive 9-th root of unity.

Then, the minimal polynomial of β has the zeros $\beta, \beta^2, \beta^4, \beta^8, \beta^{16} = \beta^7, \beta^{14} = \beta^5$.

$\Longrightarrow$ min. polynomial: $(x^9 - 1)/(x^3 - 1) = x^6 + x^3 + 1$.

So $(x^9 - 1) = \underbrace{(x-1)}_{f_1(x)}\underbrace{(x^2+x+1)}_{f_2(x)}\underbrace{(x^6+x^3+1)}_{f_3(x)}$.

The code M_3^+ has pairwise independent columns in H, ie. min.distance ≥ 3.

M_3^+ consists of codewords $(c_0c_1c_2 \;\; c_0c_1c_2 \;\; c_0c_1c_2)$, so $d = 3$.

The code M_3^- has check polynomial $x^6 + x^3 + 1$, so it's a $[9, 6]$-code.

Since $x^3 - 1 \in M_3^-$ (is a codeword), the distance is 2.

7.6 Idempotent of a cyclic code

Theorem 6.4.1. $\exists!\ c(x) \in C$ which is an identity element for C.

Proof. let $g(x)$ generator polynomial, $h(x)$ check polynomial.

ie. $g(x) \cdot h(x) = x^n - 1$ in $\mathbb{F}_q[x]$.

Since $x^n - 1$ has no multiple zeros (ie. has simple zeros), we have $(g(x), h(x)) = 1$, hence by the Bezout identity:

$$\exists\ a(x), b(x) \in F_q[x] \text{ s.th. } a(x)g(x) + b(x)h(x) = gcd(g, h) = 1$$

Define $c(x) := a(x)g(x) = 1 - b(x)h(x)$.

Clearly $c(x) \in C$. Furthermore, if $p(x)g(x) \in C$ (ie. $p(x) \cdot g(x)$ is any codeword in C), then

$$\begin{aligned} c(x)p(x)g(x) &= \underbrace{(1 - b(x)h(x))}_{c(x)}p(x)g(x) \\ &= p(x)g(x) - b(x)h(x)p(x)g(x) \\ &(recall : g(x)h(x) = x^n - 1) \\ &= p(x)g(x) - b(x)p(x)\underbrace{g(x)h(x)}_{=x^n-1} \\ &= p(x)g(x) \pmod{x^n - 1} \end{aligned}$$

So $c(x)$ is an identity element for C, and hence is unique. □

Recall:

nilpotent $a \in A$ s.th. $a^n = 0$ for $n > 0$

idempotent $e \in A$ s.th. $e^2 = e$

Definition 6.4.2 (primitive idempotent)**.** the idempotent of an irreducible cyclic code M_i^- is called a *primitive idempotent*, denoted by $\theta_i(x)$.

Theorem 6.4.3. if C_1, C_2 are cyclic codes with idempotents $c_1(x), c_2(x)$, then

i. $C_1 \cap C_2$ has idempotent $c_1(x) \cdot c_2(x)$

ii. $C_1 + C_2$, ie. the set of all words $a + b$ with $a \in C_1,\ b \in C_2$, has idempotent $c_1(x) + c_2(x) + c_1(x) \cdot c_2(x) \in C_1 + C_2$.

Theorem 6.4.4. for the primitive idempotent we have:

i. $\theta_i(x)\theta_j(x) = 0$ if $i \neq j$

ii. $\sum_{i=1}^{t} \theta_i(x) = 1$

iii. $1+\theta_{i_1}(x)+\theta_{i_2}(x)+\ldots+\theta_{i_r}(x)$ is the idempotent of the code with generator $f_{i_1}(x) \cdot f_{i_2}(x) \cdot \ldots \cdot f_{i_r}(x)$.

7.7 Trace representation

Definition 1.1.29. $q = p^r$, trace: $Tr : \mathbb{F}_q \longrightarrow \mathbb{F}_p$.
If $\zeta \in \mathbb{F}_q$, then

$$Tr(\zeta) := \zeta + \zeta^p + \zeta^{p^2} + \ldots + \zeta^{p^{r-1}} \in \mathbb{F}_p$$

Theorem 1.1.30. properties of the trace function:

i. $\forall \zeta \in \mathbb{F}_q, \quad Tr(\zeta) \in \mathbb{F}_p$

ii. there are elements $\theta \in \mathbb{F}_p$ such that $Tr(\zeta) \neq 0$

iii. Tr is a linear mapping.

Proof. i. by definition $(Tr(\zeta))^p = Tr(\zeta)$

ii. the equation $x + x^p + \ldots + x^{p^{r-1}} = 0$ can not have q roots in $\mathbb{F}_q$

iii. since $(\zeta + \eta)^p = \zeta^p + \eta^p$ and $\forall a \in \mathbb{F}_p$ we have $a^p = a$

□

Corollary . Tr takes every value $p^{-1}q$ times, and the polynomial $x+x^p+\ldots+x^{p^{r-1}}$ is a product of minimal polynomials.

Theorem 6.5.1. let k be the multiplicative order of $p mod n$, $q = p^k$, β a primitive n-th root of unity in $\mathbb{F}_q$.
Then the set

$$V := \{c(\zeta) := (Tr(\zeta),\ Tr(\zeta\beta),\ \ldots,\ Tr(\zeta\beta^{n-1})) \mid \zeta \in \mathbb{F}_q\}$$

is an $[n, k]$ irreducible code over $\mathbb{F}_p$.

Proof. by Theorem 1.1.30, V is a linear code, and $c(\zeta\beta^{-}1)$ is a cyclic shift of $c(\zeta)$; hence V is a cyclic code.

Since β is in no subfield of $\mathbb{F}_q$, we know that β is a zero of an irreducible polynomial $h(x) = h_0 + h_1x + \ldots + h_kx^k$.

If $c(\zeta) = (c_0, c_1, \ldots, c_{n-1})$, then

$$\sum_{i=0}^{k} c_i h_i = Tr(\zeta h(\beta)) = Tr(0) = 0$$

since β is a zero of $h(x)$, so that $h(\beta) = 0$.

So we observer $c(x)h(x) = 0$, from Theorem 12.14 this means that $h(x)$ is the check polynomial,
ie. we have a parity check equation for the code V.

Since $h(x)$ irreducible, $x^k \cdot h(x^{-1})$ is the check polynomial for V, therefore, V is an irreducible cyclic $[n, k]$-code. □

7.8 Mattson-Solomon Polynomial

Definition (Mattson-Solomon polynomial)**.** Let β primitive n-th root of unity in the extension field E of $\mathbb{F}_q$.

Let T the set of polynomials over E with degree at most $n-1$, ie.

$$T = \{p(x) \in E \mid deg(p(x)) \leq n-1\}$$

Define $\Phi : T \longrightarrow T$ as:
let $a(x) \in T$, then $A(X) = (\Phi_a)(X)$ is defined by

$$A(X) := \sum_{j=1}^{n} a(\beta^j) X^{n-j}$$

If $a = (a_0, a_1, \ldots, a_{n-1})$, the polynomial $A(X)$ obtained from $a_0 + a_1x + a_{n-1}x^{n-1}$ is called the *Mattson-Solomon polynomial* of the vector a.

Lemma 6.5.3. the inverse of Φ is given by

$$a(x) = n^{-1}(\Phi A)(x^{-1}) \pmod{x^n - 1}$$

Proof.

$$A(\beta^k) = \sum_{j=1}^{n}\sum_{i=0}^{n-1} a_i \beta^{ij}\beta^{-kj} = \sum_{i=0}^{n-1} a_i \sum_{j=1}^{n} \beta^{(i-k)j} = na_k$$

□

Remark . Notation:

- $\circ$: multiplication of polynomials $mod(x^n - 1)$

$*$: defined by

$$(\sum a_i x^i) * (\sum b_i x^i) := \sum a_i b_i x^i$$

Φ is an isomorphism of the ring $(T, +, \circ)$ onto the ring $(T, +, *)$.

Lemma 6.5.4. let V a cyclic code over $\mathbb{F}_q$ generated by

$$g(x) = \prod_{k \in K} (x - \beta^k)$$

Suppose $\{1, 2, \ldots, d-1\} \subset K$ and $a \in V$.

Then the degree of the Mattson-Solomon polynomial A of a is at most $n - d$.

Proof. $a(\beta^j) = 0$ for $j \in [d]$ since $g(x)|a(x)$.

Thus at the equation $A(X) := \sum_{j=1}^{n} a(\beta^j) X^{n-j}$, it will be 0 between $1 \leq j \leq d-1$, so that degree is $n - d$. □

Theorem 6.5.5. if there are r n-th roots of unity which are zeros of the Mattson-Solomon polynomial A of a word a, then $w(a) = n - r$.

Proof. consequence of Lemma 6.5.3. □

References

[1] Raymond Hill. A First Course in Coding Theory, 1986.

[2] J.H. van Lint. Introduction to Coding Theory, 1981.

[3] Pedro Tamaroff. Vandermonde determinant proof. `https://math.stackexchange.com/questions/525334/vandermonde-determinant-by-induction`.

[4] Thomas W. Judson. Abstract algebra, theory and applications. *sec.22.2.*

[5] Commutative algebra notes. `https://github.com/arnaucube/math/blob/master/commutative-algebra-notes.pdf`.